Project Executable Files

Date	15 March 2026
Team ID	
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	3 Marks

Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	Yes
5	Environment configuration file (.env.example or similar)	Yes
6	Deployed application URL (if hosted)	No
7	APK / Executable binary (if applicable for mobile/desktop apps)	No
8	Dockerfile / Containerization config (if applicable)	No
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

File / Folder Structure

```
Emotion-Detection-Learning-Support-Engine/

├── Source Code/
│   ├── app.py                   - Main application controller and entry point
│   ├── emotion_detector.py      - Emotion recognition and prediction module
│   ├── gemini_service.py        - Gemini API integration for AI-based responses
│   ├── recommendation_engine.py - Personalized learning recommendation system
│   ├── utils.py                 - Utility and helper functions
│   ├── config.py                - Application configuration settings
│   ├── requirements.txt         - Required Python libraries and dependencies
│   ├── saved_models/            - Pre-trained and saved machine learning models
│   ├── history/                 - User session and interaction history storage
│   ├── pages/                   - Frontend/UI page components
│   └── uploads/                 - User-uploaded files and media storage
│
├── README.md                    - Documentation containing project description,
│                                 installation steps, and usage instructions
```

Deployment / Access Details

Field	Details
Hosted / Deployed URL	NA
Login Credentials (Demo)	NA
Platform / Hosting Provider	NA
Repository Link	https://github.com/akshayashiny/Emotion-Detection-Learning-Support-Engine
Demo Video Link	https://docs.google.com/document/d/103PCXZD26vPzS-hgYRpgaAgOenhmrKvFjCGuzgadIgU/edit?tab=t.0

Run Instructions

Pre-requisites:

• Python 3.9 or above
• Git
• pip (Python Package Manager)
• Streamlit
• TensorFlow / Keras
• PyTorch
• Hugging Face Transformers
• NumPy
• Pandas
• Scikit-learn
• Plotly
• Google Gemini API access
• Internet connection
• Windows/Linux/macOS operating system
• Minimum 8 GB RAM recommended

Step 1: Clone the repository:
git clone https://github.com/akshayashiny/Emotion-Detection-Learning-Support-Engine.git

Step 2: Navigate to the project directory:
cd Emotion-Detection-Learning-Support-Engine

Step 3: Create a virtual environment:
python -m venv venv

Step 4: Activate the virtual environment:

Windows: venv\Scripts\activate
Linux/macOS: source venv/bin/activate

Step 5: Install project dependencies:
pip install -r "Source Code/requirements.txt"

Step 6: Navigate to the Source Code folder:
cd "Source Code"

Step 7: Run the Streamlit application:

streamlit run app.py

or

python -m streamlit run app.py

Step 8: Open the application in your web browser:

http://localhost:8501

Step 9: Use the application:

- Enter the student's learning problem.
- Select the subject.
- Choose the emotion detection model (BiLSTM or BERT).
- Click "Analyze Emotion".
- View the detected emotion and confidence score.
- Receive personalized learning recommendations.
- Generate AI guidance using Google Gemini.
- Save the session to history.
- Download the analysis report.

Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Emotion detection accuracy may vary for ambiguous, sarcastic, or very short text inputs.	Users are encouraged to provide detailed learning-related descriptions. Future work includes fine-tuning models on larger educational datasets.
2	Google Gemini AI guidance generation requires an active internet connection and a valid API key.	The application provides predefined learning recommendations when the Gemini API is unavailable.
3	The current system supports only text-based emotion detection and does not analyze voice, facial expressions, or multilingual inputs.	Planned future enhancements include speech emotion recognition, facial emotion analysis, and multi-language support.